TSOU HUNG MING (Rex)

AI Application R&D Engineer

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SUMMARY

AI Application R&D Engineer at BenQ Materials, developing and maintaining internal AI systems. Works from user needs through planning, development and production; has delivered chemical-substance management, regulatory compliance and process analytics systems, and maintains the finance and manufacturing data pipelines. Also involved in containerization, CI/CD and internal AI adoption. M.S. research in computer vision and generative models.

EXPERIENCE

BenQ Materials Corp. — Senior AI Application R&D Engineer, Taoyuan — Mar 2025 – Present

GP Chemical Substance Management Platform · built from scratch · Mar – Dec 2025

- Built a four-stage LLM compliance pipeline: PDF/Excel extraction → substance consolidation → 9-method matching with LLM adjudication → restricted-substance and threshold checks with risk grading and reports.
- Backend on FastAPI, SQLAlchemy 2.0 and SQL Server (250+ endpoints) with Azure OpenAI and Azure Document Intelligence; also serves as the session-auth and RBAC hub for the GP suite.

CosmeReg Cosmetics Regulatory Compliance · requirements to production · May 2025 – Feb 2026

- Multi-jurisdiction ingredient compliance with five-layer substance matching and SCD Type 2 regulation versioning, so past verdicts can be traced to their rule version. Saves ~800 labor hours per year.

VSM-Vision Contact Lens Process Analytics · full-stack, own AutoML design · Nov 2025 – Present

- 13 plant-wide modules including mold configuration recommendation, greedy mold scheduling, SPC/Pareto quality analysis, decision-tree root cause and gradient-boosting parameter optimization.
- Python, FastAPI, Pandas, scikit-learn; Next.js 14, React, TypeScript; offline IIS deployment.

BOSS / QVSP Finance & Manufacturing Data Systems · maintainer after handover · May 2026 – Present

- Took over 25 Airflow DAGs and 50+ tables serving BU and OU reporting; documented the systems and addressed silent-failure and non-idempotent re-run risks.
- Migrated Windows Task Scheduler jobs to Airflow, corrected revenue currency conversion, and added production-replay verification before deploys. Currently developing BOSS Phase II.

Engineering Enablement

- In-company AI adoption lecturer, sharing practical cases and applications.
- Researching Harness Engineering (LLM agent loop: plan → build → test → human review) and planning a secure internal sandbox environment.
- Researching and introducing CI/CD and Kubernetes for the department; leading internal Docker training.

SYSTEX Corporation — Full-Stack Engineer, Financial Business Unit, Taipei — Jun 2021 – Aug 2022

- **Postal audit system:** maintained the legacy VB.NET system; built an offline audit PWA with IndexedDB CRUD synchronized to the relational database.
- **Bank credit system:** developed front end, back end and API integration on .NET 6; wrote database functions, stored procedures and triggers.
- **Form and report system:** parsed Word/Excel/PDF forms and wrote results back to the database; produced test reports, vulnerability scans and user manuals.

EDUCATION

M.S., Information & Decision Sciences — National Taipei University of Business (AIoT Lab) — 2022 – 2024

B.S., Information Management — National Taipei University of Business — 2020 – 2022

PUBLICATIONS & AWARDS

- **Best Paper**, IMP 2023 — generative-AI prediction of occluded human skeleton keypoints
- **Best Paper, First Prize**, Cross-Strait Youth Sci-Tech Forum 2024 — 3D pose estimation via semantic segmentation
- IEEE ICASI 2024 — diffusion models for 2D human pose estimation under occlusion
- Excellent Paper, MC/WASN 2023 · Honorable Mention, AI Application Competition 2023

SKILLS

AI / ML: Azure OpenAI, Azure Document Intelligence, LLM pipelines, MCP, AutoML, scikit-learn, computer vision

Backend & Data: Python, FastAPI, SQLAlchemy, Pandas, Apache Airflow, SQL Server, Oracle

Frontend: Next.js, React, TypeScript, JavaScript, C# / ASP.NET MVC, .NET 6